

Semiconductor Optical Amplifier for WDM and High-Power Applications

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Abstract *Two different types of quantum-dot O-band semiconductor optical amplifiers are demonstrated using a single epitaxial structure. By varying the device length, both low-noise WDM amplification and near watt-class output power are achieved, enabling linear and high-power operation within a single technology. ©2026 The Author(s)*

Introduction

Semiconductor optical amplifiers (SOAs) operating in the O-band (1260-1360 nm) are increasingly important components for modern optical communication systems, including in datacom interconnects. Compared to fiber-based amplifiers, semiconductor devices provide a compact footprint and lower power consumption, enabling implementation in both discrete modules and integrated photonic circuits [1,2].

State-of-the-art SOAs based on quantum-dot (QD) active regions provide key advantages, including broadband gain, reduced temperature dependence, low noise figure (NF), and high reliability [3]. These properties make QD-SOAs well suited for wavelength-division multiplexing (WDM) systems, where stable multi-wavelength amplification is required. In such applications, a low NF is essential to maintain signal quality, while a combination of moderate gain and high saturation output power (OP) enables operation over a wide input signal range [1,4]. At the same time, SOAs can operate as booster optical amplifiers (BOAs) for applications requiring high OP, such as LiDAR and free-space communication systems [5].

However, achieving both low-noise amplification for WDM systems and high-power operation for BOAs remains challenging, as these regimes are typically realized using different device designs or epitaxial structures, limiting unification and increasing fabrication complexity.

In this work, we demonstrate that a single GaAs QD MBE-grown epitaxial structure can be used for both low-noise SOA operation and near watt-class BOA performance, providing a unified technological platform for distinct O-band amplification regimes.

Device design

An asymmetric layer structure consisting of a thick GaAs buffer layer, AlGaAs n-cladding, an active region containing InAs/GaAs QD, AlGaAs

p-cladding, and a GaAs contact layer, was grown with use of a production MBE setup on a GaAs (100) substrate. No overgrowth steps were applied.

The waveguide was formed by a shallow 5 μm width mesa to ensure single-mode optical confinement and a far-field profile for efficient fiber coupling. To provide natural suppression of back-reflections from the facets the mesa stripe design with 5° tilt was chosen. Additionally, both facets were covered by anti-reflection coating to ensure ripple-free operation.

The amplifier chips with lengths of 4 mm and 8 mm were cleaved from the processed structure, bonded p-side up onto AlN carriers and packaged into 14-pin fiber-coupled butterfly (BTF) modules with integrated thermoelectric cooler. Measurements were performed both ex-facet (chip-on-carrier, CoC) and ex-fiber (module level). All fabrication processes were carried out fully in-house at Innolume GmbH.

Experiment

To evaluate the amplifiers performance at the CoC level, OP as a function of current (L-I) and gain measurements were performed at 25 °C over the wavelength (WL) range 1280-1330 nm, with input optical powers of 0 dBm and 10 dBm using the setup described in [5].

The 8 mm devices were tested at currents up to 5 A, the 4 mm devices were tested up to 2 A (Fig. 1). For both devices, the OP increases with injection current and approaches gain compression saturation at higher currents. The highest OP is observed near 1310-1320 nm, corresponding to the gain maximum of the designed QD active region.

For 8 mm devices (Fig. 1(a)) shorter WL input (1300-1310 nm) exhibit higher OP at lower currents, while the gain maximum shifts toward 1320 nm due to the current-induced red-shift of the gain spectrum [6]. At a current of 4 A the OP

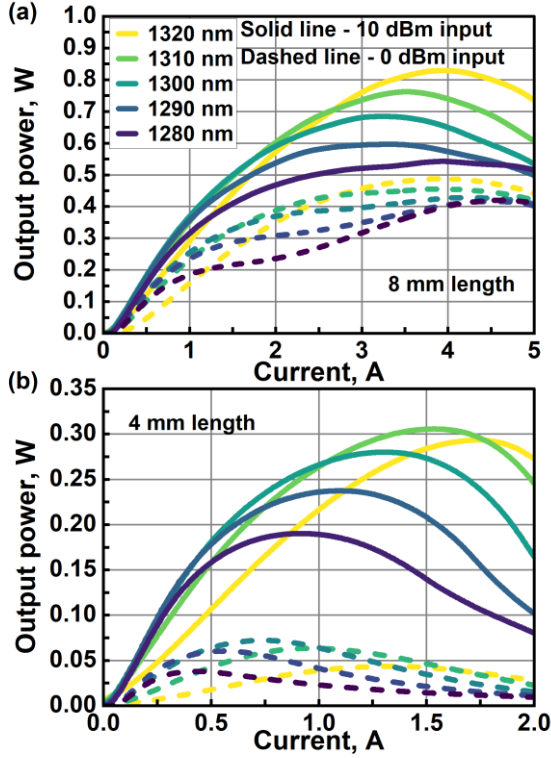


Fig. 1: Experimental results on L-I measurements of CoC at 25 °C. (a) 8 mm device. (b) 4 mm device.

exceeds 820 and 490 mW with input power of 10 and 0 dBm, respectively. The 4 mm devices (Fig. 1(b)) demonstrate similar power dependence when changing the WL. With 10 dBm input, the output signal at 1300 nm exceeds that at 1310 nm up to 1 A, afterwards 1310 nm prevail reaching a maximum OP of 310 mW at 1.55 A. For 0 dBm input, crossing of 1300 nm and 1310 nm curves is also observed around 1 A (~60 mW), but the maximum output achieved for 1300 nm is at ~0.77 A.

Fig. 2(a) depicts the amplified spontaneous emission (ASE) spectra for both 4 mm and 8 mm devices measured at 25 °C. The spectra show the two peaks corresponding to the ground state (~1300 nm) and the excited state (~1210) of the QD active region. For QD-based gain media an

increase in the excited-state contribution and decrease in ground-state contribution can be observed with an increase of current, while the overall spectrum is shifting towards longer WL.

Gain measurements were performed at 25 °C in the WL range 1270-1330 nm with input from -25 dBm to 10 dBm. Here, G_0 denotes the unsaturated gain (small-signal gain), P_{sat} is defined as the OP at the onset of saturation, corresponding to decrease of the gain by 3 dB with respect to G_0 . Having identical epitaxial structures, both devices exhibit similar gain spectral shapes, with a G_0 maximum at 1300-1310 nm (Fig. 2(b)). With increasing current, the gain peak red-shifts, following the ASE spectral shift. The 8 mm devices achieve the highest gain, with G_0 exceeding 36 dB, while the 4 mm devices reach 18-20 dB. We note that gain, measured in dB, is proportional to the chip length (linear gain is exponential with length), indicating comparable modal gain and current density conditions for 4 and 8 mm devices.

The extracted saturation output power P_{sat} in Fig. 2(c) is nearly independent of device length at the same current density, in agreement with theoretical expectations. However, for both devices P_{sat} strongly increases with current and exceeds 25 dBm for currents above 1.5 A (for 4 mm) and 3 A (for 8 mm) near 1300-1310 nm. Interestingly, P_{sat} is minimal for the wavelengths 1290-1310 nm, corresponding to the maximum gain, and increases at lower and higher wavelengths. We note that with blue and red shift of the WL, at currents above 1 A for the 4 mm device the 3 dB saturation condition could not be reached due to the limited available gain and setup input power limitations.

Fig. 3 shows the optical power performance of the fiber-coupled BTF-packaged 8 mm device. The module was mechanically fixed on a metal heatsink with thermoelectric cooling. Measurements were performed at a temperature of 25 °C, with input optical power up to 20 dBm and

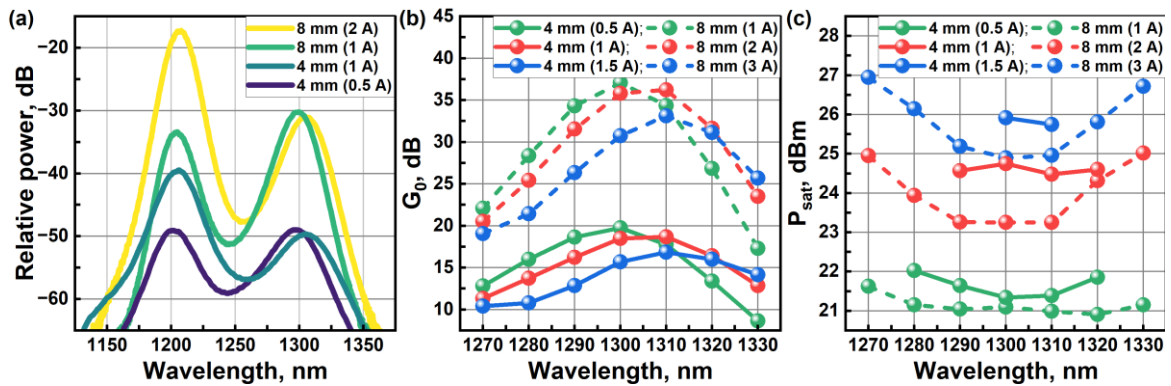


Fig. 2: Experimental results on gain performance at 25 °C. (a) Amplified spontaneous emission spectra. (b) Gain spectra. (c) Saturation output power spectra.

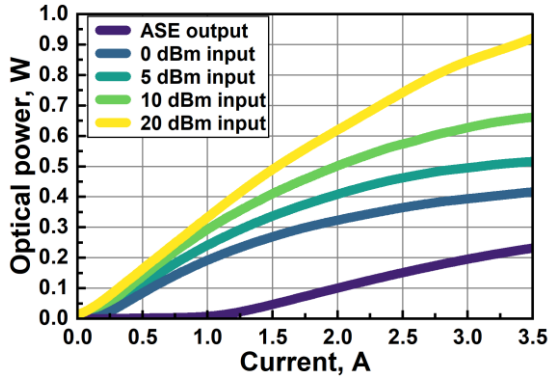


Fig. 3: Experimental results on L-I measurements of BTF-packaged 8 mm device at 25 °C.

injection current up to 3.5 A. The input WL was 1310 nm. The ASE of the device output exceeds 200 mW. With 10 dBm input power, the output exceeds 650 mW, and with 20 dBm input, the output surpasses 900 mW, illustrating that the 8 mm device operates as a high-power booster amplifier in the O-band. At higher currents, a thermal roll-over behaviour is observed. Despite the QD-based active region supporting stable and thermally robust performance [7,8], operation at high current and high optical power levels requires effective thermal management due to package-related thermal limitations. Further increase in OP is possible when operating at longer input WL (e.g., 1320 nm), consistent with the current-induced red-shift of the gain spectrum.

The NF of the 4 mm SOA is measured to be ~ 3.5 dB at CoC level and ~ 5 dB in the BTF package, indicating an additional ~ 1.5 dB loss attributed to coupling and packaging.

Fig. 4 depicts experimental result on data transmission through the 4 mm device in a BTF package using 400 Gb/s ER4 transceiver with PAM4 53 Gbaud (100 Gb/s) modulation and four LAN-WDM channels (Tx generates PRBS31). The setup is described in [9]. Spectra of the modulated optical signal before and after SOA in the Fig. 4(a) demonstrate amplification for all four optical lines with highest small-signal gain value of 15.8 dB at 1304.5 input WL.

The bit error rate (BER) (Fig. 4(b)) exhibits a characteristic U-shaped dependence on input power when the registered optical power at the receiver is fixed. We note that we depict BER, corresponding to one channel, while all 4 optical channels are transmitted through SOA simultaneously. At input power below -6 dBm the BER is relatively high due to the effect of a signal-ASE beat noise. As the input power increases, the BER decreases and reaches a minimum in the range of -2 to 3 dBm, corresponding to the optimal amplifier operating regime. At input powers above 2 dBm the BER increases again due to gain saturation and associated nonlinear distortions. For the studied operating conditions BER

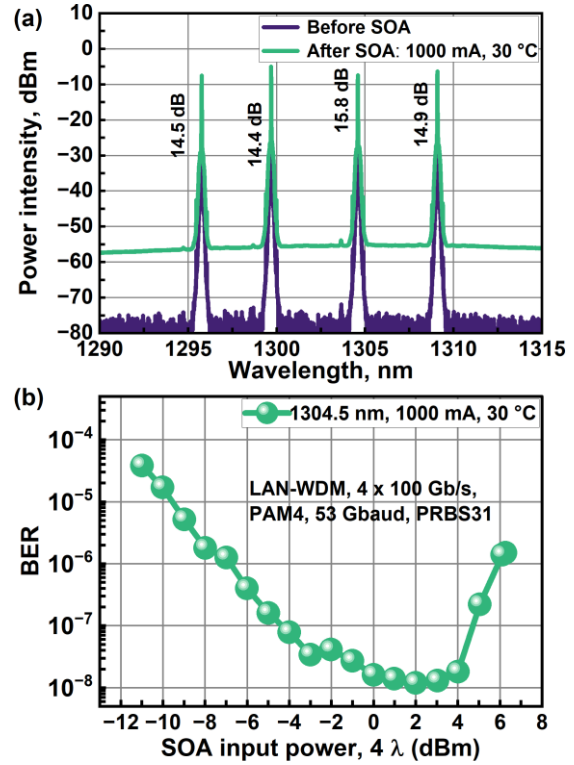


Fig. 4: Experimental results on data transmission performance of BTF-packaged 4 mm device. (a) Spectra of PAM4-modulated Lan-WDM optical signal before and after SOA. (b) BER as a function of the total SOA input power for 4 Lan-WDM channels each modulated 100 Gb/s PAM4 transmitted through SOA. The registered optical power at the receiver is the kept constant.

performance of $<10^{-6}$ is achieved within the input power range of -6 dBm to 4.5 dBm. The best performance is obtained at 1304.5 nm, 1000 mA, and 30 °C, where the BER reaches values on the order of 10^{-8} , which about 3-5 times higher than BER without SOA for the same registered optical power, but well below KP4-FEC limit (2.2×10^{-4}).

Conclusions

In this work we have demonstrated two different operating regimes using amplifiers with different chip lengths: a 4 mm device, suitable for low-noise WDM amplification, similar to fiber amplifiers such as EDFAs, and an 8 mm device, enabling high-output-power operation, for example in LiDAR applications. The 4 mm device exhibits a noise figure as low as ~ 3.5 dB at the chip-on-carrier level, and supports BER below 10^{-6} , reaching $\sim 10^{-8}$, in a BTF package. The 8 mm device delivers output power exceeding 900 mW in a BTF package. While application-specific optimization remains necessary, the results demonstrate that QD MBE-grown structures can be used as a common technological platform for O-band semiconductor optical amplifiers.

References

- [1] J. Cui, K. E. Zoiros, and A. Kotb, "Advances in semiconductor optical amplifier technologies for all-optical logic gate implementations: A comprehensive review," *Nano-materials*, vol. 16, no. 3, p. 202, 2026. DOI: [10.3390/nano16030202](https://doi.org/10.3390/nano16030202)
- [2] Y. Liu, J. Zhang, L. Bogaert, E. Soltanian, E. Delli, K. Morozov, S. Mikhlin, J. Rimböck, G. Lepage, P. Verheyen, and J. Van Campenhout, "Micro-transfer printing of O-band InAs/GaAs quantum-dot SOAs on silicon photonic integrated circuits," *Photonics Research*, vol. 13, no. 5, pp. 1341-1352, 2025. DOI: [10.1364/PRJ.545946](https://doi.org/10.1364/PRJ.545946)
- [3] V. Cao, S. Pan, D. Wu, H. Zhang, M. Tang, A. Seeds, H. Liu, X. Xiao, and S. Chen, "A novel bidirectionally operated chirped quantum-dot based semiconductor optical amplifier using a dual ground state spectrum," *APL Photonics*, vol. 9, no. 4, 2024. DOI: [10.1063/5.0194677](https://doi.org/10.1063/5.0194677)
- [4] M. Eyvazi, R. Yadipour, A. Rostami, P. Rostami, and H. Mirtaghioglu, "Theoretical analysis and design of a dual-wavelength and selectable all-optical broadband quantum-dot semiconductor optical amplifier with inhomogeneous broadening," *Optical and Quantum Electronics*, vol. 57, no. 122, 2025. DOI: [10.1007/s11082-024-08014-x](https://doi.org/10.1007/s11082-024-08014-x)
- [5] S. V. Poltavtsev, J. Rautert, V. S. Mikhlin, S. S. Mikhlin, A. E. Gubenko, and A. R. Kovsh, "High-power O-band quantum-dot booster amplifier for uncooled operation," in *Proceedings of the Optical Fiber Communication Conference (OFC)*, 2024, pp. 1-3. DOI: [10.1364/OFC.2024.Th4B.5](https://doi.org/10.1364/OFC.2024.Th4B.5)
- [6] S. Ročas, E. Vostrikov, B. Kim, V. Mikhlin, A. Zhabotinskii, J. Shapiro, S. Poltavtsev, A. Gubenko, and A. Kovsh, "Monolithically integrated O-band quantum dot DFB laser with a SOA section," in *Proceedings of the European Conference on Optical Communications (ECOC)*, 2025, pp. 1-4. DOI: [10.1109/ECOC66593.2025.11263332](https://doi.org/10.1109/ECOC66593.2025.11263332)
- [7] S.-H. Xu, D.-Y. Zhou, and L.-S. Liao, "Quantum-dot intermediate band semiconductors: From materials to applications," *Advanced Optical Materials*, vol. 13, no. 32, p. e01521, 2025. DOI: [10.1002/adom.202501521](https://doi.org/10.1002/adom.202501521)
- [8] J. Li, H. Zeng, L. Bao, J. S. Park, C. Chen, Y. Wang, H. Wang, K. Liu, Y. Zhang, M. Mtunzi, and H. Deng, "High-temperature operation of all-molecular beam epitaxy InAs/GaAs quantum dot lasers on V-grooved Si (001)," *physica status solidi (a)*, vol. 223, no. 4, p. e202501035, 2026. DOI: [10.1002/pssa.202501035](https://doi.org/10.1002/pssa.202501035)
- [9] V. V. Belykh, J. Rautert, V. S. Mikhlin, and A. E. Gubenko, "Bit error rate in WDM data transmission links with semiconductor optical amplifier," *Journal of Light-wave Technology*, vol. 43, no. 3, pp. 1076-1088, 2024. DOI: [10.1109/JLT.2024.3475885](https://doi.org/10.1109/JLT.2024.3475885)